# Apologies, Empathy, and Persistence: How Conversational Tutors' Affective Responses Shape Learner Effort

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## Abstract

When a learner expresses frustration, a conversational tutor can ignore it, apologize, or acknowledge the feeling and reframe the difficulty. In a Wizard-of-Oz study with 96 participants solving logic puzzles, we compared these three response styles. Acknowledge-and-reframe increased the number of puzzles attempted after a failure by 28% relative to ignoring, while apologizing had no effect on persistence and lowered perceived tutor competence. Participants described scripted empathy as hollow when it repeated.

# 1 Introduction

Affect-aware tutors often respond to detected frustration with sympathy. Whether sympathy helps learners keep going is less clear.

# 2 Study

96 participants solved a sequence of logic puzzles with a chat tutor operated by a trained wizard following scripts. Frustration was induced with one unsolvable puzzle. Participants were randomly assigned to one of three response styles, triggered when they expressed frustration in text.

| | Example response | Puzzles attempted after failure | Perceived competenc |
|---|---|---|---|
| | continues with next hint | 3.2 | 5.1 |
| gize | "Sorry, that one was hard." | 3.3 | 4.4 |
| wledge + reframe | "That's frustrating. People usually get stuck here because ..." | 4.1 | 5.6 |

# 3 Qualitative findings

Participants found acknowledgement useful when it explained why the task was hard. The same sentence repeated a second time was described as "robotic" by 19 participants. Apologies were read as the tutor admitting it had made a mistake.

# 4 Design implications

• Pair any acknowledgement of feeling with information about the task.

• Vary wording and avoid repeating the same empathic phrase.

• Avoid apologies unless the tutor actually erred.

# 5 Limitations

Puzzles, not course material; a single induced failure; a wizard rather than an automated detector.

# References

[1] Picard, R. W. Affective Computing. MIT Press, 1997.

[2] Albrecht, S. and Nwosu, C. Frustration markers in text-based tutoring dialogues. IJAIED 32, 4 (2022), 901-930.